Customer Churn Intelligence Platform

An end-to-end data and machine learning platform for identifying customers at risk of churn

The Problem

Telecom companies have large volumes of customer, service and billing data, but identifying **which customers are likely to leave** is difficult.

Challenges:

- Customer information is spread across multiple data layers.
- Manual rule-based identification does not capture complex patterns.
- Not every high-risk customer is obvious from a single attribute.
- Retention teams need **prioritized customers**, not just raw data.
- Predictions must be accessible through an application, not only a notebook.

Business Question

“Which customers are most likely to churn, and who should the retention team prioritize?”

The Solution: Customer Churn Intelligence Platform

An end-to-end platform that transforms raw customer data into **actionable churn risk intelligence**.

What it does

1. Ingest
Collect and validate customer data.

2. Clean & Transform
Standardize data and handle missing/invalid values.

3. Engineer Features
Create meaningful behavioral and customer-level features.

4. Analyze
Identify high-risk customer segments.

5. Predict
Use ML to calculate individual churn probability.

6. Visualize
Expose insights through APIs and a React dashboard.

Outcome: From raw customer records → prioritized retention actions

Product Workflow

Customer Data

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File Ingestion

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Schema & Data Quality Validation

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Data Cleaning

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Feature Engineering

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MySQL Curated Data

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ML Model

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Customer Risk Score

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FastAPI

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React Dashboard

Product Features

Customer Search

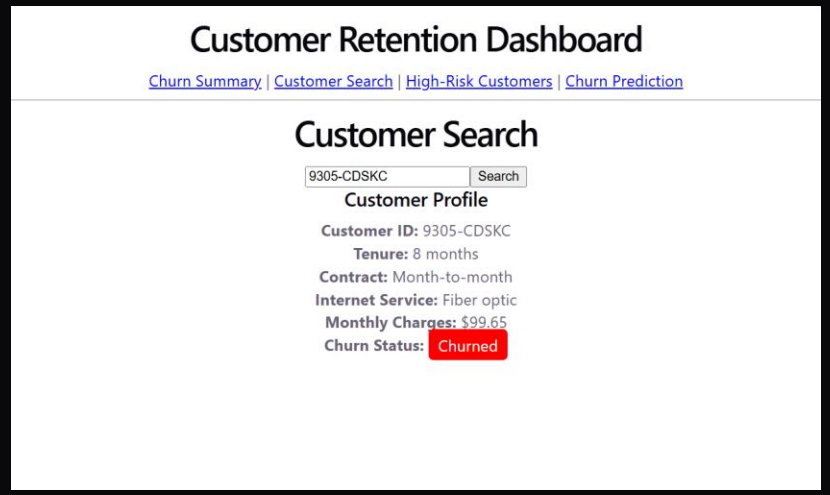
Search for an individual customer and view:

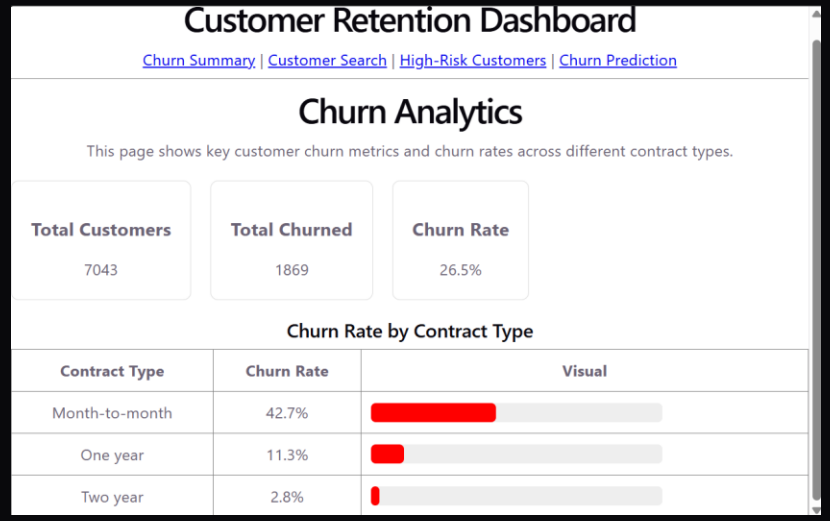
- Tenure
- Contract
- Internet service
- Monthly charges
- Current churn status

Churn Analytics

View:

- Total customers
- Total churned
- Overall churn rate
- Churn rate by contract
- Churn rate by internet service





Product Features

High-Risk Customers

View customers identified by business rules with:

- Risk reason
- Contract
- Tenure
- Monthly charges

Churn Prediction

Enter customer information and receive:

- Risk score
- Prediction
- Confidence
- Visual risk indicator



